- **Concept of Triple Integral, Triple integral in Cartesian, Cylindrical and Spherical Coordinate system.**

## Triple Integrals

**READ:**

**READ: HOWARD ANTON: 14.5: Triple Integrals**

**WATCH:**

**Calculus - How to find the bounds of a triple integral**

**Triple Integrals in Cartesian Coordinates | Volume between Surfaces**

**PRACTISE: HOWARD ANTON: 14.5: Triple Integrals: 1-12, 15-18**

**1–8** Evaluate the iterated integral. ■

**1.** $\int_{-1}^{1}\int_{0}^{2}\int_{0}^{1}(x^2+y^2+z^2)\,dx\,dy\,dz$

**2.** $\int_{1/3}^{1/2}\int_{0}^{\pi}\int_{0}^{1} zx\sin xy\,dz\,dy\,dx$

**3.** $\int_{0}^{2}\int_{-1}^{y^2}\int_{-1}^{z} yz\,dx\,dz\,dy$

**4.** $\int_{0}^{\pi/4}\int_{0}^{1}\int_{0}^{x^2} x\cos y\,dz\,dx\,dy$

**5.** $\int_{0}^{3}\int_{0}^{\sqrt{9-z^2}}\int_{0}^{x} xy\,dy\,dx\,dz$

**6.** $\int_{1}^{3}\int_{x}^{x^2}\int_{0}^{\ln z} xe^{y}\,dy\,dz\,dx$

**7.** $\int_{0}^{2}\int_{0}^{\sqrt{4-x^2}}\int_{-5+x^2+y^2}^{3-x^2-y^2} x\,dz\,dy\,dx$

**8.** $\int_{1}^{2}\int_{z}^{2}\int_{0}^{\sqrt{3}y}\frac{y}{x^2+y^2}\,dx\,dy\,dz$

**9–12** Evaluate the triple integral. ■

**9.** $\iiint_G xy \sin yz \, dV$, where $G$ is the rectangular box defined by the inequalities $0 \le x \le \pi, 0 \le y \le 1, 0 \le z \le \pi/6$.

**10.** $\iiint_G y \, dV$, where $G$ is the solid enclosed by the plane $z = y$, the $xy$-plane, and the parabolic cylinder $y = 1 - x^2$.

**11.** $\iiint_G xyz \, dV$, where $G$ is the solid in the first octant that is bounded by the parabolic cylinder $z = 2 - x^2$ and the planes $z = 0$, $y = x$, and $y = 0$.

**12.** $\iiint_G \cos(z/y) \, dV$, where $G$ is the solid defined by the inequalities $\pi/6 \le y \le \pi/2, y \le x \le \pi/2, 0 \le z \le xy$.

**15–18** Use a triple integral to find the volume of the solid. ■

**15.** The solid in the first octant bounded by the coordinate planes and the plane $3x + 6y + 4z = 12$.

**16.** The solid bounded by the surface $z = \sqrt{y}$ and the planes $x + y = 1$, $x = 0$, and $z = 0$.

**17.** The solid bounded by the surface $y = x^2$ and the planes $y + z = 4$ and $z = 0$.

**18.** The wedge in the first octant that is cut from the solid cylinder $y^2 + z^2 \leq 1$ by the planes $y = x$ and $x = 0$.

**Triple Integrals in Cylindrical and Spherical Coordinates**

**READ:**

**READ: HOWARD ANTON: 14.6: Triple Integrals in Cylindrical and Spherical Coordinates**

**WATCH:**

**Integration in Spherical Coordinates**

**PRACTISE: HOWARD ANTON: 14.6: Triple Integrals in Cylindrical and Spherical Coordinates: 9-12, 13-16, 17-20**

**9–12** Use cylindrical coordinates to find the volume of the solid.

**9.** The solid enclosed by the paraboloid $z = x^2 + y^2$ and the plane $z = 9$.

**10.** The solid that is bounded above by the sphere $x^2 + y^2 + z^2 = 1$ and below by the cone $z = \sqrt{x^2 + y^2}$.

**11.** The solid that is inside the surface $r^2 + z^2 = 20$ but not above the surface $z = r^2$.

**12.** The solid enclosed between the cone $z = (hr)/a$ and the plane $z = h$.

**13–16** Use spherical coordinates to find the volume of the solid.

**13.** The solid bounded above by the sphere $\rho = 4$ and below by the cone $\phi = \pi/3$.

**14.** The solid within the cone $\phi = \pi/4$ and between the spheres $\rho = 1$ and $\rho = 2$.

**15.** The solid enclosed by the sphere $x^2 + y^2 + z^2 = 4a^2$ and the planes $z = 0$ and $z = a$.

**16.** The solid within the sphere $x^2 + y^2 + z^2 = 9$, outside the cone $z = \sqrt{x^2 + y^2}$, and above the $xy$-plane.

**17–20** Use cylindrical or spherical coordinates to evaluate the integral. ■

**17.** $\displaystyle\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{a^2-x^2-y^2} x^2\,dz\,dy\,dx \quad (a > 0)$

**18.** $\displaystyle\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} e^{-(x^2+y^2+z^2)^{3/2}}\,dz\,dy\,dx$

**19.** $\displaystyle\int_0^2 \int_0^{\sqrt{4-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{8-x^2-y^2}} z^2\,dz\,dx\,dy$

**20.** $\displaystyle\int_{-3}^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} \int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2}\,dz\,dx\,dy$